

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0099

Calamus 100 kV
March 30, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0099. The Interconnection Customer proposes a 114.8 MW natural gas combined cycle facility interconnecting at the Calamus 100 kV substation in the Ohio area. The request was received on January 24, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 88 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,225,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The POMARIA / ST MARYS 100.0 kV is overloaded to 113.7% of its 134 MVA emergency rating for the loss of the EAST METHUEN / SWAN VALLEY 500.0 kV L. The project contributes 6.0% DFAX (6.9 MW) to this overload.

The LENKURT / SOUTHWEST LIMA 100.0 kV line is overloaded to 111.5% of its 153 MVA emergency rating for the loss of the EASTOWN RD 230/1 kV Auto. The project contributes 6.3% DFAX (7.2 MW) to this overload.

For the loss of EASTOWN ROAD 230 kV/1 kV Transformer, the GEISEL MONUMENT / BAVERLEY 100.0 kV line loads to 111.2% of its 135 MVA rating. Project DFAX: 6.3%.

For the loss of EAST METHUEN / SWAN VLY SUBSTATION 500.0 kV line, the TAIBAN MESA / POMARIA 100.0 kV loads to 108.9% of its 130 MVA rating. Project DFAX: 6.0%.

The BEAVERLEY SUBSTATION / SOUTHWEST LIMA 100.0 kV Line is overloaded to 107.6% of its 149 MVA emergency rating for the loss of the EASTOWN RD 230/1kV Autotransformer. The project contributes 6.3% DFAX (7.2 MW) to this overload.

The TWIN COUNTY RANCH / SWAN VALLEY 230.0 kV is overloaded to 106.3% of its 632 MVA emergency rating for the loss of the SWAN VLY / TREGO 500.0 kV. The project contributes 11.6% DFAX (13.4 MW) to this overload.

The ST MARYS / EAST DRAINAGE CANAL 100.0 kV Circuit 2 is overloaded to 104.5% of its 205 MVA emergency rating for the loss of the E DRAINAGE CANAL / ST MARYS 100.0 kV Cir. 1. The project contributes 5.3% DFAX (6.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ST MARYS / POMARIA 100.0 kV	Rebuild line with higher-capacity conductor	\$2,287,000	100% (\$2,287,000)
SOUTHWEST LIMA / LENKURT SUBSTATION 100.0 kV	Rebuild line with higher-capacity conductor	\$5,772,000	100% (\$5,772,000)
GEISEL MONUMENT / BAVERLEY 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,992,000	100% (\$1,992,000)
TAIBAN MESA / POMARIA 100.0 kV	Reconductor line	\$4,012,000	100% (\$4,012,000)
BEAVERLEY SUBSTATION / SOUTHWEST LIMA 100.0 kV Line	Reconductor line	\$792,000	100% (\$792,000)
TWIN COUNTY RANCH / SWAN VALLEY 230.0 kV	Reconductor line	\$18,973,000	11% (\$2,092,000)
ST MARYS / EAST DRAINAGE CANAL 100.0 kV Circuit 2	Reconductor line	\$278,000	100% (\$278,000)
	Total allocated to project		\$17,225,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.